Cyber-attacks on cameras in the IoT networks

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***Abstract*— The use of IoT devices has been a ubiquitous phenomenon nowadays, different functionalities are being designed, manufactured, and connected to the network so they can now interconnect and exchange data with each other. Despite the benefits, these devices are also subject to several security risks. This paper will present an overview that targeting some common cyber-attacks of internet-connected surveillance camera in IoT networks, comparing threats and available solutions to discuss the problems and future research directions.**

Keywords- Cyber-Attacks; Cyber-Security; Internet of Things; Network security; Surveillance camera.

I. INTRODUCTION

Due to the popularity of the internet, the living standards of most users have improved considerably. Users now can access the latest information and news through the internet; make long video calls with friends and family even they are in different countries; use the internet to shop online to purchase products from thousands of miles away that are not available locally, the internet has made all these actions feasible and easy. And now, as people's living standard and needs expand, the world has become more connected than ever before, and a large part of this has to do with the Internet of Things.

Internet of Things (IoT) technology has been adopted worldwide in recent years as billions of devices are connected to the network, which allows them to exchange and communicate information without human supervision and operation, and this data can later be collected and used for more analytical needs. Not only big events or large areas like city constructions and large enterprises use IoT technology, but also individual users are using IoT technology to make their life easier and more convenient.

IoT devices can connect to the internet through wired or wireless connections and then transmit information through sensor units. The IoT network can host many devices within the network. With home appliances such as smart light bulbs or wearable devices forming a smart home network architecture, or smart medical devices and industrial devices forming an industrial IoT (IIoT), it is reasonable to assume that more and more tools or devices will be able to connect to the IoT network in the future.

Among all types of smart homewares, smart surveillance cameras account took up a significant proportion of the total number of IoT devices, according to (Abdalla & Varol, 2020), with almost a 108million internet-enabled cameras being sent to buyers around the world to serve a variety of purposes. However, a significant amount of private information has been exposed due to the recent attacks on surveillance cameras.

The role of surveillance cameras has become more diverse, from monitoring and recording large scale areas with restricted access, such as inside factories or residential area, even urban traffic monitoring; surveillance cameras are now designed to be more versatile and intelligent. The characteristics of these cameras now require them to be placed in locations where private areas can be photographed and because of user's lack of awareness of the devices allows some unscrupulous attackers to take advantage of them. Security cameras are supposed to be used for the safety and protection of users, but due to cyber-attacks, they have become one of the main methods to attack on users' privacy. At the same time due to these threats more people are choosing not to use security cameras, but this is not the most effective solution and goes against the reason why they need them in the first place. There are some solutions that researchers purposed such as strengthened penetration test and authentication processes, but as protection method are enhanced, attackers are also working on more approaches to compromise the devices at the same time.

This paper will investigate the types and methods of attacks that internet-enabled surveillance cameras may be exposed to in the IoT network; an introduction to webcams; the types of crimes committed against webcams and the measures that could be taken to protect surveillance cameras. Finally, concludes with a discussion and summary through an analysis of the relevant literature.

II. TYPES OF CYBER-ATTACKS IN IOT

The impact of cyber-attacks on traditional devices such as computers or mobile phones has declined as the enhanced security features and the users' safety awareness improved. However, more IoT devices are being purchased and added to the network as shown in figure 2, people are spending more money each year on IoT devices and expected to be more in the future. Meanwhile, smart home devices are taken big part of it and due to the competitive and fast-paced nature of the IoT homeware devices industry, a wide range of products have been manufactured that do not or just meet the minimum requirement of network security standards. Many manufacturers use cheap, low-quality components to achieve a competitive price to gain market shares, but the use of cheap hardware and the lack of security features makes it difficult for most users to detect and repair any security breaches or attacks promptly that might cause serious data loss. As a result of low computing power, limited security protection mechanism and other several main reasons, cybercriminals are now targeting IoT devices, and the following are different types of attack that are frequently occurred in the IoT network.

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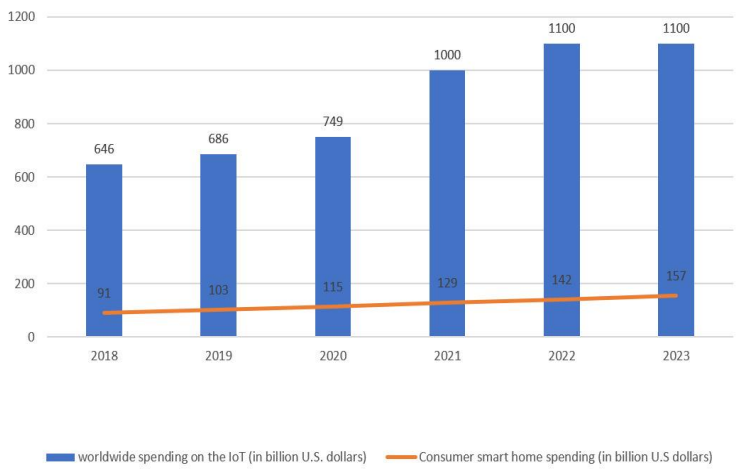

Fig. 1. Forcast worldwide spending on IoT and smart home spending in 2018 - 2023

A. *Physical Attack*

Attacks are coming from not only inside but also from outside, due to the limitations of the IoT devices protection. (Deogirikar & Vidhate, 2017) Comparing to other forms of attack methods, the methods of physical attack are common and varied. Not all devices can be installed in a secure location where they cannot be manipulated by anyone other than the user, such as some outdoor surveillance cameras: even though storage technology is constantly evolving, some cameras are still designed with internal storage, where a secure digital (SD) memory card is inserted inside the machine to back up the content, this allows criminals to physically damage the device to gain access to the stored user's personal data. Although more IoT devices now support cloud storage, criminals can still manipulate the nodes where data is transferred to gain access to user privacy. It is even possible to interfere with RFIDs by manipulating the radio frequency to make information exchange difficult, which is also a form of DoS attack.

B. *DoS/DDoS Attack*

The Denial of Service (DoS) is one of the most frequently used attack methods by attackers to attack on the availability of devices or services in a network, to cause the target to be overtaxed on hardware or network bandwidth resources in a constant state of being attacked so they unable to function or provide services. Example attack for DoS could be a Ping of Death attack, by keep sending large number of oversized packets to the destination address with ping command through command prompt which could be done on every computer that is connected to the network, oversized packet would usually be fragmented into small pieces, so when receiver attempt to reassemble the packet would not success and keep writing to the memory cause further overflow to exhaust victim device resources.

1) *Distributed Denial of Service (DDoS) Attack*

Distributed Denial of Service (DDoS) differs from DoS attacks in it is not constrained by the attacker's network bandwidth and is a distributed attack on the target, using the resources from others to continuously send large numbers of packets to the target causing network congestion and resource consumption on the target device for attacking. An example of DDoS could be SynFlood attacks, attackers initiate the attack by sending a mass of SYN packets to the victim, then when the receivers try to respond to each one in sequence, by default it will have a port in open status to receive the follow ACK packets. When countless syn packets are received, the victim side continues to process the syn packets and keep multiple ports open, over time, resources get exhausted on the victim side of the device and makes it difficult to serve for its purposes.

a) *Botnet Attack*

Botnet attacks, another attack method widely used to target IoT devices. Botnets primarily manipulate zombie hosts that are controlled by malware or virus infections and are often used in DDoS attack as a part of the crime tool. If a hostmaster targets a victim, a large number of botnet devices would send out an unexpected amount of harmful information, causing network congestion and blocking ports so that regular transmission of information or services cannot be performed, and each victim's device is used as a zombie machine to attack more devices. In recent years, IoT devices have become the main target for botnet attacks due to the massive uses of IoT devices. According to reports, by 2020 there were nearly 8.7 billion IoT devices connected to the network worldwide, and most of these devices only equipped with a low level of protection but they do have enough processing power to be used for attacking.

Mirai is a well-known botnet attack method that exploits vulnerabilities in IoT devices. This method is constantly attempting to login to a device port in the users' network with common usernames and passwords, and then automatically executing a series of shellcodes to download the mirai trojan. It became well-known because of the series of attacks conducted during 2016 (Idriss, 2020), attacker raided DNS provider DYN Company caused serious loss for services providers like Amazon, Twitter, Spotify etc. Meanwhile, due to the source code leaking issue, hackers had built numbers of different variants with other exploits.

b) *Jamming Attack*

Due to the characteristic of most IoT devices are connected through wireless connections, attackers targeting victim device system by maliciously interfering with the regular transmission of information by the frequent occupation of the communication channel, so that packets have to be transmitted repeatedly despite the difficulty of reaching their destination with high possibility result in victim device's battery drained (Sunil Ghildiyal, 2014).

III. ATTACKS ON IOT CAMERAS

However, these issues were rarely addressed until recent years, more and more attacks have shifted from targeting large corporate IoT network or devices to small area use or personal devices, and this is directly attack on the individual user's privacy.

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A. *Types of Camera*

Most camera people are using in the current society could be one of following:

Cameras	Characteristics	Transmission type
Digital Video Recorder (DVR)	Hard disk storage; Compressed data for easy access; Limited deployment options.	Coax Cables
Network Video Recorder (NVR)	Front-end connect with network camera; Higher resolution than DVR.	Ethernet Cables
IP Camera (IPC)	Function independently; Web-based operating system; Direct connect with internet.	Wireless connection

Fig. 2. Different camera types used in the market.

B. *IoT cameras*

Baby monitors - which are used to keep track of the child when the parent needs to leave the child for a short time, and allow the child to be seen in real-time by the parent and most of them have an intercom function.

Security cameras - placed in the living room or at the entrance for security purposes, usually with motion detection and infra-red capabilities for 24/7 surveillance, and some devices equipped with AI face recognition can provide identity tagging to alert stranger access.

Talkable cameras - allow users to have two-way conversations with elderly people, young children even pets stay at home by themselves, provide function of 360-degree monitoring of the home and communicating with them in case of any health issue or emergencies happen for helps.

All these network-connected cameras could be connected with other IoT devices, such as google home player or iPhone, so devices can be controlled through voice assistant function.

C. *Attacks and Threats on Network Connected Camera*

The nature of these domestic cameras means that they can be installed and used in intimate spaces, such as bedrooms and living rooms etc. And it is this feature that has led to multiple webcam break-ins all around the world.

The Washington Post reported a shocking story in December 2019 (Chiu, 2019) about a mother of three children, who installed the famous brand Ring family camera in her kid's room to monitor their activities and safety anytime. What happened afterwards was shocking, as a strange male voice appeared through the talkable camera attempting to lure and scare the children. The parents believe the incident has caused great harm and privacy breaches, as they have no idea of when the hacker started accessing their surveillance equipment and viewing their children's private behaviors including sleeping and changing clothes. Also, they can't track down the hackers and the incident has caused a traumatic experience for the family. However, this is not a single incident of a Ring device being attacked, as another woman in Georgia, USA, was verbally abused by a hacker while she was sleeping, leaving her terrified. (Francis & Landrigan, 2019)

According to ABC news (Roberts, 2020), countless IoT cameras around the world have been hacked and the images they capture are being used for webcasting. Website called Insecam based in Russia is live streaming images being captured by countless IoT cameras, and the variety of camera brands that have been hacked has proved that camera brand is not the only reason for the attacks to be successful.

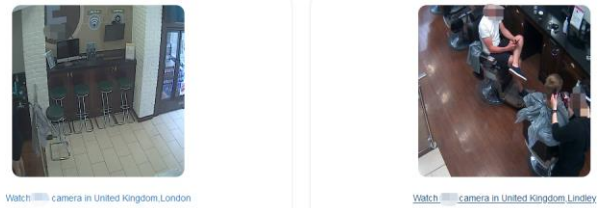

Fig. 3. Hacked cameras been streaming online.

And the attack against Verkada, a leading US security camera company, in March this year once again drew the world's attention to the issue of camera security. As a well-known video security company, Verkada sells security cameras and services that enable customers to access and manage them remotely. The main attraction of their products is the artificial intelligence (AI) facial recognition technology application, which is used in a range of corresponding security services. Several large international companies, even hospitals, schools and prisons are using the services they offer. Therefore, there were nearly 150,000 security cameras got hacked in this recent malicious attack. Hackers posted captures of the hacked camera including the Tesla assemble line, a prison's canteen, gym entry access record and more; meanwhile the hackers claim they are gaining access to not only the security cameras but also all the profiles that each facility is having. If their claim is verified, then there would be an immeasurable amount of data compromised in this attack. This could a testimony to the ultra-large range of surveillance by IoT cameras and the insecurity of IoT device systems today.

IV. PROTECTION METHODS.

The following would discuss different measures that can take place to improve the security of these cameras.

A. *Security mechanism protection*

1) *Unsupervised Algorithm in Machine Learning*

Machine learning is one of the best approaches in this scenario, as its ability to perform unsupervised learning makes it highly probable in dynamic networks. Camera devices in IoT networks often have limited data processing and energy supply capabilities, leading to a disadvantage in case of external attacks. With the application of machine learning, it is possible to learn how to identify attacks based on existing attack datasets, thus having the ability to quickly identify the danger and provide a timely solution when an attack starts.

However, the current limitations of ML for solving security problems also arise about the collection of data sets: to react to attacks in the shortest time possible. The algorithm makes the devices regularly learn from a real-time dataset of attacking data in the network, which can be inaccurate without human

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detection leading to incorrect learning results and further problems if the database is maliciously contaminated by an attacker. (Gopal et al., 2021)

B. Manufactory Protection

There are even more options for manufacture to secure the product's security:

a) Hardware Protection

There is an increasing number of products are designed with a camera such as a robot vacuum cleaner or a smart refrigerator etc. besides the protection mechanism on the system what manufacturers can also do is to reduce the unnecessary insecurity, such as a sliding camera cover, instead of leaving the camera on when the device is activated, users could manually cover it as needed.

b) Security Penetration Test

According to (Abdalla & Varol, 2020), a security penetration test is aimed to find the potential risks and flaws, to conduct a series of actions including black-box, white-box and grey-box testing for detecting firmware, applications, and connectivity.

c) Service Protection

Most IoT devices could be remotely accessed with application on a mobile device, multiple authentications could be used to ensure security. For example, when an IoT device is connected to the network using the software for the first time, it will need to require the user's account and Bluetooth to complete the connection, which will restrict the accessibility of unknown devices. Meanwhile, when users setting up their new password, complexity should be required as well.

C. User Protection

Insecam on their website homepage claimed that most IoT cameras are streaming on this website because the devices are still using a default password which has not been changed, if anyone wish to remove their camera from there simply by resetting a new password for their devices. Professional analyzed mirai botnet source code and to get the result it is developed based on more than 60 combinations of username and weak password including the later developed variants Masuta and Okiru were taking advantage of it. (Idriss, 2020) To mitigate by change the default password with stronger ones could greatly help to reduce the possibility of being attack.

V. SUMMARIZATION

Current research shows the dangers and uncertainties in today's fast-growing IoT networks and in this article a short overview of the various types of cyber-attacks that exist today such as DoS/DDoS attacks. Device manufacturers and developers should not only focus on developing innovative products but also on securing the safety of users. Evidence showed that an enormous amount of illegal private information is being sold or traded through the internet. Without proper security mechanisms in place, there will be no privacy for the world that already surrounded by cameras, whether they are children in their own homes, patients in hospitals or prisoners in prisons.

As part of my future work, I will focus on working out if there is a pre-installed system or software that can be used on small home IoT devices to prevent and detect cyber-attacks.

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